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# Charging Ahead: Prepaid Metering, Electricity Use, and Utility Revenue†

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*Monthly bills for services such as electricity often go unpaid in developing countries. Prepaid meters offer a potential technological solution. In Cape Town, South Africa, over 4,000 residential customers on monthly billing were switched to prepaid metering, with random variation in the timing of the switch. In response, electricity use falls by 14 percent, driven at least in part by an increase in marginal price sensitivity. The decrease in revenue to the municipal electric utility is more than offset by lower revenue recovery costs. Switching poorer and more in-debt customers generates the greatest net revenue gains to the utility.* (*JEL* L94, L98, O13, Q41, Q48)

The use of lights measured from space to proxy for economic activity (Henderson, Storeygard, and Weil 2011) drives home the importance of electricity for development. Much of the global increase in electricity demand in the coming decades is forecast to come from developing countries (Wolfram, Shelef, and Gertler 2012). However, on the ground, maintaining a reliable supply of affordable electricity presents challenges for both service providers and households. The standard model for recovering the costs of electricity provision—like the provision of many other public and private services—is a postpaid billing system in which a customer is charged for consumption over the past month. For poor, liquidity-constrained customers, finding resources to pay the bill presents a major challenge. At the same time, enforcing bill payment is costly to the utility and politically infeasible in many settings. Kojima and Trimble (2016) estimate that the total annual value of uncollected electricity bills is 0.17 percent of national GDP on average in sub-Saharan Africa. This narrative echoes difficulties with revenue collection for many public services in developing countries.

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Prepaid electricity meters offer a technological solution to the nonpayment problem. These meters, which operate on a debit basis, are increasingly common in both developed and developing countries, yet little is known about their effect on consumption or on revenue recovery and costs. We implemented a field experiment in collaboration with the municipal electric utility in Cape Town, South Africa, in which over 4,000 residential customers were switched from monthly billing to prepaid metering. Following the metering switch, electricity use fell by around 14 percent. We showed that the impact on the electric utility's bottom line is positive overall, with the greatest gains coming from the types of customers who utilities—and other types of service providers in both the public and private sectors—are least willing to serve when revenue must be recovered ex post.[1]

Given these findings, the rise of prepaid electricity metering is perhaps unsurprising. For example, in the United States, utilities in 34 states offer some type of prepayment plan for electricity, where they are used both to address nonpayment of bills and as a demand-side management strategy (Drehobl 2017). The greatest expansion in prepaid metering, however, is happening in the developing world, where new electrification is fueling demand for the meters. Market forecasts indicate that prepaid meters will make up the majority of the electricity meter market in Africa by 2020 (Northeast Group 2016).[2] South Africa was an early adopter of prepaid electricity metering during the phase of rapid electrification following the end of apartheid, when electricity was rolled out to poor and rural communities (Gaunt 2005, Bekker et al. 2008, Dinkelman 2011). In spite of growing interest in prepaid metering to improve revenue recovery, we are unaware of any prior causal estimates of the effect on customer behavior in a developing country, or analysis of whether prepayment does, in fact, improve the financial viability of serving low-income customers.

We generate new evidence on the impact of prepaid metering on two main sets of outcomes. First, we ask how transitioning customers from postpaid monthly billing to prepaid metering affects electricity use. Unlike the impacts on payment behavior, which are largely mechanical, the impacts of prepaid metering on consumption are hard to anticipate based solely on customer behavior on a postpaid meter.[3] Second, we examine the implications for the electric utility's bottom line, which is affected

[1] Regulatory restrictions often prevent public service providers from actively barring customers from access. Electric utilities have been shown to respond by worsening the quality of infrastructure in less profitable neighborhoods (McRae 2015b) or restricting access through high connection fees (Golumbeanu and Barnes 2013; Lee, Miguel, and Wolfram 2016; Lee et al. 2016; Blimpo, McRae, and Steinbuks 2018). On many tariff structures, customers using small amounts of electricity each month are more costly to supply for reasons that go beyond the risk of nonpayment. First, if recurring costs such as meter reading and bill processing are recovered out of consumption charges, then low levels of consumption are less likely to cover these costs. Second, on increasing block tariffs, low consumption results in a lower average marginal price for electricity.

[2] The expansion of prepaid metering is not limited to Africa. The Northeast Group (2016) estimates that an additional 126 million prepaid meters will be rolled out between 2016 and 2026, with growth in all parts of the world. See Section IA for additional detail.

[3] Specifically, postpaid metering mechanically results in payments after consumption occurs, while a switch to prepaid metering automatically moves payment to a date earlier than consumption. Observing payment delinquency on postpaid metering, and assuming that all prepaid payments will occur immediately before consumption, therefore provides a realistic lower bound on the impacts on payment timing, with bigger effects should customers choose to purchase further in advance. Similarly, prepaid metering automatically eliminates payment defaults; observing default rates on postpaid metering again offers a reasonable benchmark for the impacts of prepaid metering on overall revenue recovery rates (ignoring potential impacts on illegal consumption). Consumption, on the

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not only by the change in consumer demand but also by differences in the timing, completeness, and cost of revenue recovery across metering types. We observe monthly electricity use and payment behavior for the 4,175 residential customers included in a mandatory meter replacement project in Cape Town. The phase-in of the meter replacements was randomly ordered across 27 geographic areas over a six-month window. We use the randomized phase-in order to instrument for prepaid metering status in each month of the phase-in window, which recovers a causal estimate of the impacts of prepaid metering that avoids selection into metering type. Prepaid metering affects numerous aspects of consumption, and we lack the data to estimate impacts on consumer welfare. We focus, instead, on estimating consumption and payment responses as an input to the electric utility's revenue and cost calculations. We are particularly interested in heterogeneity in the effects on the utility's net revenue, which has implications for a utility's willingness to serve otherwise unprofitable customer types.

We find that customers reduce their electricity use when switched from postpaid monthly billing to prepaid electricity metering by 1.9 kWh per customer per day, or around 14 percent.[4] The reduction is stable in the months following meter replacement, and is robust to a number of alternative specifications. The largest reductions come from high consumers, who also tend to face a higher marginal price, and from poorer customers.

We speculate that a number of mechanisms may underlie the changes in consumption that we observe. First, prepaid meters enforce payments automatically and so increase the experienced price of electricity relative to postpaid metering (even though tariffs remain the same), particularly where enforcement is lax. Second, by forcing customers to pay in advance, prepaid metering may affect expenditure patterns, particularly for credit-constrained customers. Third, the change from postpaid to prepaid metering transfers some transaction costs from the utility (meter reading, billing) to the customer (purchasing, monitoring consumption). Finally, a number of informational and behavioral differences characterize the switch. For example, in-home displays provide feedback on usage, more frequent purchases make expenditures more salient, and the prepaid system offers a form of self-control and intrahousehold control.

These channels have different implications for consumer welfare. They are not mutually exclusive and our study cannot cleanly distinguish between them. Instead, we provide empirical support for one potentially important channel, price information and salience. Electricity in Cape Town is priced on an increasing block tariff for both postpaid and prepaid customers, with prepaid customers moving up the tariff schedule based on cumulative purchases during the month. As prepaid customers increase the frequency with which they purchase electricity, average and marginal prices at the transaction level converge. Greater sensitivity to price need not, therefore, come from a more sophisticated understanding of the increasing block tariff.

other hand, might increase or decrease when customers are switched to prepaid metering, affecting both the revenue from and cost of electricity supply.

[4] For comparison, the reductions associated with nonprice demand-side management strategies are around 1–2 percent (e.g., Allcott 2011). Our results are more consistent with reductions under critical peak pricing, during which prices rise by up to 600 percent (Wolak 2011; Jessoe and Rapson 2014).

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Instead, high frequency price feedback makes the higher marginal price at higher levels of consumption (within the month) more salient. Consistent with this mechanism, we observe bunching of total monthly usage around the largest tariff step for prepaid customers, who purchase electricity and receive price feedback every three days on average, and no bunching for postpaid customers. We follow the public finance literature to quantify the magnitude of the bunching (see Kleven 2016). Prior to the meter replacement project, the bunching elasticity is 0.01 (standard error 0.02); following the switch, it is −0.12 (standard error 0.04). While greater sensitivity to marginal prices is consistent with lower optimization frictions on the part of customers, we hesitate to draw conclusions about the impacts of prepaid metering on consumer welfare, and leave it as a topic for future research.[5]

The estimated decrease in consumption results in a corresponding decrease in the revenue due to the utility of around US$6.90 per customer per month, and it saves the utility around US$3.10 in average monthly kWh supply costs.[6] However, more complete revenue recovery, lower recovery costs, and payments that arrive around three months earlier, on average, tip the balance in favor of prepaid electricity meters from the utility's perspective. Specifically, the value to the utility of moving payment earlier is equal to around US$1.10 per customer per month, while the value of eliminating nonpayment altogether is US$6.40 per customer per month. Thus, even in a setting where the overall likelihood of nonpayment is low, relative to other parts of Africa (Trimble et al. 2016), the value of complete revenue recovery is a substantial component of the gains to the utility from prepaid metering. In our setting, the additional net revenue from a prepaid meter makes up for the higher fixed cost of the meter in less than a year. Accounting for changes in electricity use is important: assuming customer behavior remains fixed increases the projected returns from prepaid metering (relative to postpaid billing) by almost 50 percent.

Together, our results indicate that prepaid electricity metering can help overcome revenue recovery challenges, particularly for the types of customers who are least profitable to the electric utility on a monthly billing model (i.e., poor customers who consume little power and are often delinquent on their bills), and therefore generate the highest returns from the switch to prepaid metering. While these customers benefit from lenient enforcement of bill payment, they also generate an externality on other customers by undermining the revenue base necessary for infrastructure expansion and maintenance. Prepaid metering makes substantial progress in narrowing the gap in net revenue to the utility across different types of customers in our sample, with implications for expanding electricity access in other settings. While our point estimates may not generalize, the heterogeneity that we observe suggests that prepaid metering will be more beneficial to the electric utility in settings with a more delinquent customer base or a smaller margin per kWh than Cape Town. Our

[5] Greater sensitivity to marginal prices at the transaction level that ignores the underlying tariff structure also results in misoptimization, with too much consumption on lower price tiers. Other concerns about the impact of prepaid meters on consumer welfare include, but are not limited to, the fact that liquidity-constrained customers may spend time with zero credit on their meter and therefore no electricity in their homes, as well as the transaction costs associated with purchasing electricity from shops and kiosks, which may impose an additional burden on elderly or disabled households.

[6] All values are reported in 2014 real values, based on an exchange rate of 11.45 ZAR/USD.

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setting allows us to focus on the impacts of prepaid metering in an environment free from confounds associated with new connections or unreliable supply. In addition, we do not observe how illegal consumption responds to prepaid electricity metering, which has the potential to undo the benefits to the electric utility if theft increases dramatically.

Our paper contributes to broad questions in development economics about how to overcome weak institutions that undermine revenue collection and the delivery of public services. A growing number of empirical studies on taxation show that increasing information disclosure or changing the incentives associated with enforcement improves revenue recovery (Kumler, Verhoogen, and Frías 2013; Carrillo, Pomeranz, and Singhal 2014; Pomeranz 2015; Khan, Khwaja, and Olken 2016).[7] Our results echo the conclusions in the taxation literature that shifting the enforcement burden onto the payee may improve revenue collection even in settings where monitoring and enforcement might otherwise be difficult. More specifically, our study joins a growing number of papers showcasing the potential for technology to substitute for or augment better incentives for public sector workers or beneficiaries (e.g., Duflo, Hanna, and Ryan 2012; Banerjee et al. 2016; Muralidharan, Niehaus, and Sukhtankar 2016). We also contribute to a small but growing body of literature on water and electricity billing and payment in developing countries. For example, in a recent paper, McRae (2015a) documents the heterogeneous impacts of metering (as opposed to a fixed monthly fee) on household welfare and utility revenue in Colombia. In a study of water bill payment in South Africa, Szabó and Ujhelyi (2015) show that an information intervention increased bill payment rates.

Our findings offer the first evidence on the impact of prepaid metering in a developing country context, where they are rapidly becoming the standard technology for residential connections for both on-grid and off-grid electricity. Existing work on the effects of prepaid electricity metering is scarce and consists largely of descriptive studies (Tewari and Shah 2003; Baptista 2013). A paper on the largest prepaid metering program in the United States, the Salt River Project, finds reductions in consumption of around 12 percent after customers voluntarily switch to prepaid metering (Qiu, Xing, and Wang 2017). The authors rely on a matching design to compare prepaid and postpaid customers and do not calculate payoffs to the utility. We are not aware of any other plausibly causal evidence on the effects of prepayment for electricity.[8]

Our results on the impacts of prepayment on consumer behavior have implications beyond the utility sector. Mobile phones, for example, have expanded dramatically across the developing world in recent decades due, in part, to prepayment technology that allows users who lack the credit history required for a monthly contract to still access the network (Aker and Mbiti 2010; Kalba 2008). Both monthly billing and prepayment plans coexist in most mobile phone networks, yet no empirical

[7] A parallel literature on the delivery of public services focuses on selection and incentives for public agents (e.g., Muralidharan and Sundararaman 2011; Dal Bó, Finan, and Rossi 2013; Ashraf, Bandiera, and Lee 2018) and community involvement in allocation and monitoring (e.g., Björkman and Svensson 2009; Alatas et al. 2012), among other mechanisms for improving service delivery.

[8] Gans, Alberini, and Longo (2013) study the effect of a change in the meter interface that provides prepayment customers in Ireland with additional feedback about real-time usage.

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work has directly compared their effect on phone use or other outcomes. The distinction between prepayment and postpayment also relates to a broad literature on the use of different payment modalities (debit versus credit cards, for example) and their effect on consumer behavior.[9] We view our results as evidence that the timing of payment relative to consumption has a substantial causal effect on consumption, though further work is needed to isolate the channels of impact.

The paper proceeds as follows. In the next section, we provide background on prepaid electricity meters and on the study setting. Section II describes the dataset and the empirical strategy, Section III presents the empirical results, Section IV investigates mechanisms, and Section V calculates the costs and benefits to the utility. The final section concludes.

## I. Background and Context

We begin with a general description of prepaid electricity metering before turning to the specifics of our setting, including electricity tariffs and billing in the City of Cape Town, and details of the meter replacement project that we study.

### A. *Prepaid Electricity Meters*

Prepaid electricity meters work on a debit basis: customers purchase electricity and load it on to their meter. As long as the meter has a positive balance, current flows through the meter into the home. Once the balance reaches zero, the current is interrupted. Prepaid meters display the number of units (kWh) remaining on the meter, and many have features to inform the customer when the balance is getting low, such as colored or blinking lights or an audible alert. Prepaid meters are also generally located within the dwelling in a visible and easy-to-reach place. Contrast this with what we refer to as postpaid metering. On a conventional postpaid meter, customers consume electricity and periodically have their meter read by an employee of the electric utility. Bills are sent based on cumulative consumption as recorded by the meter reader, and customers typically have an additional grace period before their bill is due. The physical meters used in most postpaid systems are located outside of the home and display consumption in a way that is difficult for the consumer to access and understand.[10]

Prepaid electricity metering is attractive to the electric utility for a few reasons. First, it generates revenue in advance of consumption and cuts down on nonpayment or late payment of electricity bills, though the latter depends on how tightly bill payment is enforced on a postpaid system. Second, it eliminates the need to send

[9] A considerable literature in behavioral economics and marketing examines how the coupling of purchase and consumption decisions affects behavior, yet relatively little field evidence exists to corroborate the laboratory and descriptive findings (see, for example, Prelec and Loewenstein 1998; Soman 2001; DellaVigna and Malmendier 2006).

[10] Recent innovations with in-home displays and dynamic pricing have begun to change the information feedback on some postpaid systems (e.g., Jessoe and Rapson 2014). Smart metering systems offer some of the same benefits as prepaid meters, such as remote disconnections and automating meter reading. They tend to be more expensive than prepaid meters (roughly three times the hardware cost alone), however, and rely on more sophisticated grid and communication infrastructure. For these reasons, smart meters are still far less common than prepaid meters in developing countries (Northeast Group 2016).

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meter readers to physically inspect meters or implement disconnections, and therefore addresses shirking or bribery in addition to labor costs and safety concerns. Third, it eliminates preparing and mailing monthly bills and processing incoming payments. These last two benefits to the utility are achieved by transferring some of the transaction costs to the customer. The prepaid system is not without costs to the electric utility, however. There is a substantial initial cost of developing a vending network to track and charge customers. Vendors typically earn a commission on sales of prepaid electricity, which may be passed on to the consumer or deducted from revenue to the utility (the latter in our setting). Prepaid electricity meters may also affect theft if customers shift from nonpayment to illegal connections.

Customers purchase prepaid electricity from physical or electronic vendors, including supermarkets, small shops and kiosks, ATMs, gas stations, and online or via mobile phone. On the vending system used in Cape Town, customers provide a meter number and the monetary value or number of kWh that they would like to purchase. The vending system issues an encrypted, meter-specific code that the customer enters into a keypad on the physical meter. The meter itself does not communicate directly with the grid.[11] See online Appendix Figure A.1 for an example of receipts from Cape Town. Over time, prepaid metering technology has improved both in terms of reliability and theft prevention. The prepaid meters that we study are known as split prepaid meters because the actual meter is located outside the home in a locked kiosk, with only the display and keypad inside of the home. Communication between the meter and the in-home display is by wire or radio frequency. This design minimizes the risk of tampering and allows the utility to perform maintenance more easily.

Prepaid metering is expanding across the developing world. Many countries in Africa, Latin America, and South Asia have established either prepaid electricity services or ongoing large-scale pilots (see Esteves et al. 2016 for an overview). Market forecasts indicate that an additional 126 million prepaid electricity meters will be rolled out between 2016 and 2026, with investments totaling US$11 billion. In Africa, prepaid meters are expected to constitute the majority of the electricity metering market by 2020 (Northeast Group 2016). Prepaid metering is already widespread in South Africa, Rwanda, and Mozambique; Ghana, Nigeria, Kenya, Uganda, Zambia, Ethiopia, and Angola are rapidly expanding prepaid meter coverage (Esteves et al. 2016; Taale and Kyeremeh 2016). Outside of Africa, China has a significant number of households on prepaid electricity metering (Du, Guo, and Wei 2017), India and Bangladesh have large-scale pilot projects underway (Esteves et al. 2016), and Indonesia is expanding the use of prepayment particularly through new construction (Nugroho et al. 2017). In Latin America, Brazil, Argentina, Colombia, Peru, Argentina, Haiti, and the Dominican Republic are all increasing use of prepaid meters (Esteves et al. 2016). Prepaid electricity meters are also found in developed countries, including New Zealand, Australia, Ireland, and the United Kingdom, all

[11] This description covers the most common type of prepaid metering systems at this point in time. Earlier generations of prepaid meters were coin-operated or relied on a physical card or key. The technology used by the City of Cape Town is the STS system, developed in South Africa in the 1980s. It is currently used for most prepaid systems around the world.

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of which have established prepaid metering programs. In the United States, utilities in 34 states offer prepaid programs (Drehobl 2017).[12]

### B. *Electricity in Cape Town, South Africa*

In 1990, as South Africa began the transition to democracy, less than a third of South Africans had access to electricity. By the end of that decade, the figure had doubled (Bekker et al. 2008), and by 2011, roughly 80 percent of South African households were electrified (IEA 2013). This extremely rapid expansion of electrification was facilitated in part by the introduction of prepaid electricity metering, which helped manage revenue recovery for previously unelectrified households (Bekker et al. 2008). In the City of Cape Town, where we focus, electrification rates in formal settlements are over 99 percent.

The national state-owned utility, Eskom, owns and operates most generation sources, as well as the national grid. It sells power in bulk to municipalities, including Cape Town, which pay time-of-use rates that vary by time of day and month of the year. The City of Cape Town supplies power to roughly 80 percent of residents (the rest are supplied by Eskom), around 450,000 (75 percent) of whom are on prepaid metering.

The City of Cape Town did not charge a fixed-service fee in the years we study, so the increasing block tariffs are set to cover both fixed and variable costs of electricity supply.[13] The tariffs charged on the prepaid meter are the same as on postpaid metering. On a prepaid meter, customers move up the tariff blocks based on cumulative purchases during the calendar month; the tariff resets on the first of each month. Two tariffs are used in most of the study years. Residential customers who consume below a threshold quantity of electricity in a 12-month rolling window are on what is referred to as a "Lifeline" tariff, which provides free electricity for up to the first 60 kWh of consumption in a calendar month. Customers not on the Lifeline tariff are charged on a comparatively flat increasing block tariff ("Domestic" tariff). Online Appendix Figure A.2 shows the tariffs for 2014–2015 in 2014 US dollars. Tariffs are updated each July.

Customers of the City of Cape Town historically received individual bills for each service (water, electricity, refuse removal, etc.). Over the last decade, customers have been shifted to a consolidated billing model, which includes all utilities on a single bill. In our sample, roughly two-thirds of customers received a consolidated bill prior to the switch to prepaid metering. A sample consolidated bill is shown in online Appendix Figure A.3. We discuss the implications of consolidated billing for our results in subsequent sections.

[12] We are grateful to Reuven Sussman of the ACEEE for guidance on the coverage of prepaid metering.

[13] In other words, the average marginal price exceeds the average marginal cost, and the difference is used to cover maintenance, new infrastructure, and other fixed operating costs. This also means that lowering consumption will reduce revenue more quickly than it reduces costs.

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### C. *Meter Replacement Project*

In late 2014, the City of Cape Town initiated a project to replace postpaid meters with prepaid meters in selected areas. Suburbs with a high penetration of prepaid meters and a low average property value were targeted, with the idea that eliminating the final few postpaid meters from these areas would cut out entire meter reading routes.[14] The project consisted of a pilot followed by two phases of implementation. Phase 1 targeted 2,240 postpaid meters in a single suburb called Mitchells Plain between November 2014 and February 2015, with successful replacement of over 90 percent of the targeted meters. Phase 2 targeted 1,935 postpaid meters spread across 14 different parts of the city between February and April 2015. Replacement rates were lower in the second phase because compliance became voluntary beginning in April 2015.

The meter replacement project proceeded as follows. A contractor, hired to complete the meter replacements, worked in geographically contiguous groups of customers identified by the City of Cape Town. Based on customer addresses, the contractor first delivered notices to targeted customers informing them that they would have their postpaid meter replaced with a prepaid meter. The project was not widely publicized, so this notice was the first time most customers learned of the project. Customers were instructed to call the contractor to schedule an appointment. The letter described a time window for scheduling and informed customers that if they had not scheduled an appointment within 15 days, their electricity would be disconnected. This window was eventually extended for an additional 15 days. When a customer called, the scheduling window available to them was determined by the randomized order of their geographic group and by contractor availability. Customers were disconnected from electricity for a couple of hours, at most, while their individual meters were replaced. Most meter replacements in a group occurred over a few days.

Figure 1 summarizes implementation. The randomization groups are ordered on the vertical axis, with the Mitchells Plain groups (Phase 1) corresponding to groups 1–13. The solid circles indicate the date on which individuals in a group first learned about the project. The triangles indicate the median switch date among those who ever switched, with the share of customers who ever switched printed on the figure. We discuss how this process is used in our empirical strategy in Section IID.

### D. *Neighborhood Characteristics*

Our sample comes from the suburb of Mitchells Plain and 14 other areas in Cape Town (see online Appendix Figure A.4 for a map of the study area). We describe the characteristics of the neighborhoods in our study, based on the 2011

[14] Though we do not observe the reasons that customers switch to prepaid metering prior to the project, reasons include moves, debt write-offs, new construction, and self selection. The customers remaining on postpaid billing in project neighborhoods may not be representative, and it is not obvious whether they are likely to be more or less sensitive to their metering technology than customers that were switched earlier.

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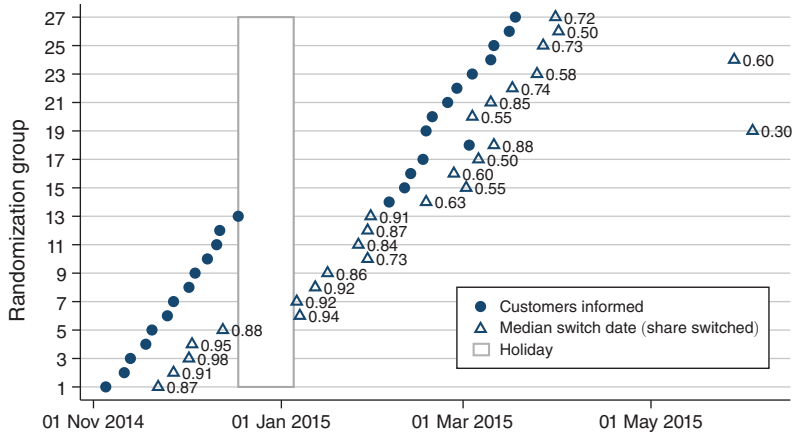


Figure 1. Switching Patterns by Randomization Group

*Notes:* Project implementation by randomization group (groups 1–13 correspond to the Mitchells Plain sample). Customers were first informed of the project via letters dropped at their homes. The median switch date is conditional on switching, and the share of customers who ever switched in the group is printed next to the marker.

South African census.[15] We begin by describing the Phase 1 sample in Mitchells Plain, a lower- to middle-income neighborhood in the Cape Flats area of Cape Town. During apartheid it was designated as a colored area, and residents were largely excluded from high-paying jobs and received less access to education as a result of apartheid policies.[16] In 2011, average monthly income was less than US$300 for 42 percent of Mitchells Plain residents, which was close to the average for the City of Cape Town. The median household spent 8–10 percent of its monthly income on electricity. The unemployment rate among working-age adults was around 32.5 percent, which was higher than the city average. Electrification was nearly universal: 99 percent of the 67,993 households in the 2011 census used electricity for lighting. Nearly all households (95 percent) in Mitchells Plain lived in formal dwellings, and owner occupancy rates were high (78 percent). Customers in Phase 2 of the project are similar to those in Mitchells Plain in most dimensions. In 2011, the other areas were, on average, slightly poorer than Mitchells Plain: unemployment was 35 percent in 2011, and 51 percent of households had monthly incomes below US$300.

Electricity use in the project sample is high relative to households in other developing countries (see Section IIC for summary statistics). Here we describe patterns in the census that help explain high electricity use, focusing on Mitchells Plain. Households rely on electricity for most of their energy needs: 95 percent of households use electricity for cooking, 85 percent for heating, and 99 percent for lighting. Ownership of energy-consuming durables is also high: 94 percent of households

[15] Unless otherwise indicated, all figures are the authors' own calculations from Statistics South Africa's Census 2011 Community Profile datasets, the small area layer GIS dataset for this census, and the publicly accessible interactive data provided by Statistics South Africa.

[16] "Colored" is a designation for people of mixed ethnic origin in the South African census.

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own a refrigerator, 95 percent own a television, and 91 percent own a cell phone. Together with a large average household size of 4.6 individuals, these patterns help explain high electricity usage rates in the study sample.

## II. Data, Study Design, and Empirical Strategy

### A. *Data*

We obtain data from the City of Cape Town's billing system and prepaid vending system under a nondisclosure agreement. Here, we summarize key features of the data. Online Appendix C contains further details on construction of the dataset and key variables.

*Billing Data*.—The City of Cape Town follows a consolidated billing model for many postpaid customers (65 percent of our sample at the time of the project) and provides a single bill that covers electricity, water, refuse, sewerage, property taxes, and debts (see online Appendix Figure A.3 for a sample bill). Bills are sent to customers approximately every 30 days, with payment due 25 days after the posting date. Electricity charges are based on physical meter readings taken every 25 to 35 days. We use meter reading dates to construct an average daily kWh variable that assumes a constant rate of consumption per day between meter readings. We then take the average across all days in the month, some of which may have come from different bills. We also construct a measure of the corresponding amount owed for consumption in each month, based on the customer's tariff.

The bill also indicates the date and value of any payments made since the last bill. We use this information to construct measures of the days after consumption that a bill is paid off and an indicator for late payments. We construct three other measures of delinquency. First, we calculate the share of all monthly bills prior to the start of the meter replacement project that were paid past the due date. Second, we construct an indicator for whether the customer had multiple outstanding bills at the end of the panel. Third, we construct a measure of whether a customer was ever disconnected prior to the project and include the cost to the city of disconnecting and reconnecting the customer in our cost-benefit analysis.

*Prepaid Vending Data*.—The prepaid vending system records electricity purchases. We use transaction dates to construct an average daily kWh variable that assumes a constant rate of consumption per day between transactions. We then take the average across all days in the month, some of which may have come from different transactions. This averaging assumes that customers are not accumulating electricity credit on their meter. As with the postpaid billing data, we also construct a measure of the corresponding amount owed in the month, based on the customer's tariff. We calculate the time between consumption and payment (which will be $\leq 0$) using the purchase date and the midpoint of the time (in months) before the next purchase.

*Average Marginal Cost of Supply and Revenue Recovery Costs*.—We obtain data on the average marginal cost of electricity supply from the City of Cape Town.

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The city purchases electricity from Eskom, the national provider, at tariffs that vary by time of day and month of the year.[17] We use the city-wide monthly average price per kWh paid to Eskom as our average marginal cost of supply measure (see online Appendix Figure A.5). The electricity department has an estimated loss rate of 11.25 percent. This includes both technical and nontechnical losses, which we assume affect both meter types equally. Prepaid electricity vendors receive approximately US$0.002 per kWh sold, or around 1.5 percent of the average marginal price observed in our data, which is deducted from the revenue remitted to the city. We also gather information from the electricity department on other administrative costs associated with revenue recovery on postpaid billing, including the average per-customer meter reading and bill preparation costs.

*Electricity Outcomes across Metering Types*.—The billing data and prepaid vending data measure electricity use in different ways. As described above, the billing data are used to construct a daily average between bill $t$ and $t-1$, while the prepaid dataset is used to construct a comparable measure between transaction $t$ and $t+1$. The constructed average daily consumption per month removes differences in the frequency of observations across meter types.

Given that the prepaid metering system records transactions rather than actual consumption, the comparability of the measures across systems deserves further discussion. A key assumption in transforming expenditures into a measure of average daily consumption is that customers are not accumulating unused electricity on their meter. Three observations suggest this is reasonable. First, the increasing block tariff structure provides an incentive not to accumulate unused electricity since tariffs reset on the first of the month, as discussed above. Second, customers tend to purchase very small amounts of electricity, very frequently: the median purchase size is 19 kWh or around 30 ZAR (US$2.60), and the median purchase frequency is every 3 days (see Jack and Smith 2015 for further discussion of purchasing patterns in this context). Finally, while the prepaid technology might lead to greater smoothing if customers have a tight monthly budget constraint, we note that the within-customer coefficient of variation of our average daily kWh measure is very similar for postpaid and prepaid observations (0.24 for postpaid and 0.21 for prepaid). The assumption that customers are not accumulating unused electricity on their meter may be most difficult to satisfy in the month or two following the meter replacement, as customers build up a minimum balance on their meter. Our main analysis drops the month of meter replacement.

More generally, the transformation of expenditure data into a measure of consumption is a common challenge in studies of household demand. In the case of prepaid electricity, we construct a monthly measure of consumption based on a mean (median) purchase frequency of 12.6 (9.5) times per month. Over 95 percent of customers make at least one purchase every month on prepaid metering. While a

[17] Roughly 45 percent of Cape Town's electricity purchases from Eskom are charged on time of use tariffs, with peak prices that are three to four times nonpeak prices during the high demand season and around 50 percent higher than nonpeak prices during the low demand season. Most of the seasonal variation is associated with peak prices. The other 55 percent of purchases are not charged on time of use, so prices vary only by month of the year. We use the monthly record of the average cents per kWh paid to Eskom provided to us by the City of Cape Town.

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number of econometric methods have been advanced for estimating latent demand based on less frequent purchases (e.g., Kay, Keen, and Morris 1984; Blundell and Meghir 1987), the high frequency of observed expenditures together with the administrative nature of our data allows us to make the simplifying assumption that monthly expenditures and monthly consumption converge.

### B. *Study Design and Implementation*

We collaborated with the City of Cape Town to randomize the phase in of the meter installation project. Groups of around 150 adjacent customers were randomized in two separate waves, one for each of the meter replacement phases. First, in Mitchells Plain (Phase 1), we imposed a grid over a map of all targeted customers and aggregated customers into cells until around 150 adjacent customers formed a randomization group. Thirteen groups were identified and their order was randomized. Second, targeted customers in the other suburbs (Phase 2) were aggregated in a similar process, but using preexisting suburb definitions to assign customers to groups. The second stage included 14 groups, which were also randomly ordered. Thus, the final design consists of 27 groups, with separate random assignment within the first and second phases. A map of the randomization groups is shown in online Appendix Figure A.4.

*Potential Confounds*.—Three details of implementation present potential confounds. We discuss these in turn, anticipating the robustness checks in Section IIIA. First, when the city installs a new meter, it is defaulted onto the Domestic tariff. If the meter should be assigned to Lifeline tariff, the change has to be manually entered into the billing or vending system. Because of the volume of meter replacements around the time of the project, there were some delays in restoring the Lifeline tariff to customers who had received it prior to the switch. Specifically, 923 of the 1,294 customers on Lifeline in the month before the switch were put on to the Domestic tariff for at least a portion of the month following the switch. The majority of these cases were corrected within three months of the switch; however, some were delayed and appear to be permanent changes. Anecdotally, few customers noticed the change. A tariff change that coincides with the switch to prepaid metering presents an obvious confound to identifying the effects of the meter on electricity use. We control for erroneous tariffs in all of our analyses, and we drop these customers altogether in a robustness check. Our analysis of heterogeneous treatment effects also shows separate results for Domestic tariff customers who were unaffected by the tariff error.

Second, as described above, around two-thirds of customers receive a consolidated bill for all utilities (electricity, water, and sewerage) prior to the meter replacement project. As a result, some of the effects of the meter replacement project may be due to the transition away from consolidated billing, which may, for example, raise the salience of electricity costs independent of the metering technology. In practice, differences in the postpaid experience on a consolidated bill versus a separate electricity bill is minimal; the consolidated bill contains all of the information about consumption, tariffs, and total amount owed for electricity that is provided under separate billing. The types of customers who receive separate bills prior to

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the project are different from those on consolidated billing: they use less electricity, have older accounts with the city, and are less likely to be located in Mitchells Plain (Phase 1). The response of these customers to prepaid metering may therefore differ from the sample average for numerous reasons. Nonetheless, we implement a robustness check that omits customers on consolidated billing prior to the meter replacement project.

Third, around the same time as the meter replacement project, the City of Cape Town implemented several months of rolling blackouts (load shedding) to manage supply shortages. These involuntary outages may lead to a reduction in electricity use that is unrelated to prepaid metering if they are correlated with the randomized timing of the prepaid meter phase in. This is unlikely to affect our estimates given that load shedding rotates across neighborhoods in Cape Town according to a prespecified schedule that affects customers at relatively high frequency (every one to two days) for short intervals, with every neighborhood in the city included in the rotation.

### C. *Study Sample*

The randomized meter phase in affects 4,175 customers who we observe over a four-and-a-half-year period, from January 2012 through June 2016. We focus our analysis on the six-month meter phase-in window between November 2014 and April 2015; summary statistics corresponding to the phase-in window are described in Table 1. The phase-in window includes a mean of 5.2 observations per customer, 2.7 of which are from a prepaid meter. Preproject customer characteristics (panel B) are constructed based on observations that precede the start of the project in November 2014. On average, customers use around 16 kWh of electricity per day prior to the project and owe around US$52 per month for electricity. Over half of their bills are paid late, and it takes an average of around three months to pay after the date of consumption. Twenty-six percent of customers have multiple outstanding unpaid bills at the end of the panel, and 20 percent were ever disconnected as a means of enforcing bill payment. The median property value is around US$27,000, and 31 percent of customers are on the subsidized tariff at the time of the project.

Relative to other customers in the neighborhoods involved in the study (the project affected less than 10 percent of customers in targeted neighborhoods), the project sample is less likely to be on a Lifeline tariff and uses slightly more electricity on average prior to the project. Once on a prepaid meter, project customers consume roughly 0.8 kWh per day more than customers not targeted by the project, conditional on being on a Lifeline tariff, and around 1 kWh per day less, conditional on being on a Domestic tariff.

### D. *Empirical Strategy*

We observe electricity and other billing outcomes for a customer $i$ in randomization group $g$ in month $t$. In the absence of any selection into prepaid metering,

$$y_{igt} = \alpha + \beta prepaid_{it} + \tau_t + \eta_i + \epsilon_{igt} \tag{1}$$

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Table 1—Sample Statistics

| | Number (1) | Mean (2) | Median (3) | SD (4) | Min (5) | Max (6) |
|---|---|---|---|---|---|---|
| *Panel A. Phase-in window observations* | | | | | | |
| Total observations | 21,759 | — | — | — | — | — |
| Total customers | 4,175 | — | — | — | — | — |
| Observations per customer | — | 5.21 | 5 | 1.414 | 1 | 6 |
| Prepaid observations per customer | — | 2.73 | 0 | 1.095 | 0 | 4 |
| *Panel B. Preproject customer-level statistics* | | | | | | |
| Daily kWh | — | 16.21 | 15 | 8.15 | 0 | 101 |
| Amount owed per month | — | 52 | 49 | 35 | 0 | 404 |
| Months late | — | 3.19 | 2 | 4.31 | −32 | 46 |
| Share paid late | — | 0.54 | 0.58 | 0.33 | 0 | 1 |
| Outstanding debts | — | 0.26 | — | — | 0 | 1 |
| Ever disconnected | — | 0.20 | — | — | 0 | 1 |
| Lifeline tariff | — | 0.31 | — | — | 0 | 1 |
| Property value | — | 29,975 | 26,662 | 14,222 | 10,704 | 156,908 |

*Notes:* Preproject customer-level statistics are calculated for all observations before November 2014. Months late is the time in months between when consumption occurs and payment is received by the utility. Share paid late is the share of months in which the customer paid past the billing due date. Outstanding debts is an indicator for whether the customer had multiple unpaid bills at the end of the panel. Ever disconnected equals one if payment was ever enforced through disconnection. Lifeline tarrif is the share of customers that had ever received the Lifeline tariff prior to the project. Property values are assessed values for tax purposes. All monetary units are in 2014 US dollars.

identifies the impact $\beta$ of prepaid metering on outcome $y_{igt}$, where $prepaid_{it}$ indicates whether customer $i$ received electricity through a prepaid meter in month $t$, $\tau_t$ are month fixed effects, and $\eta_i$ are customer fixed effects.[18] Our main results exclude the switch month, resulting in a binary, time-varying measure of $prepaid_{it}$. Standard errors are clustered at the randomization group level. In our main analyses, we account for the small number of clusters (observations $= 27$) by reporting $p$-values for standard errors estimated by wild-bootstrap (Cameron, Gelbach, and Miller 2008).

While eligibility for the meter replacement was randomly assigned, the actual meter replacement date may not be exogenous to electricity consumption. First, there is a nontrivial amount of noncompliance, particularly in Phase 2 of the project. Second, the timing of replacement even within the assigned windows may be determined by factors correlated with electricity use, such as contractor work schedules or resistance to the replacement. In exceptional circumstances, customers were allowed to make appointments outside of the assigned replacement window. Third, in practice, the length of time that the contractor spent in each group was determined by the difficulty in scheduling all appointments.

The randomized phase-in study design provides a source of exogenous variation that we use to construct instruments for the endogenous $prepaid_{it}$ variable. We split the randomized phase-in order, $g$, by the two implementation phases (groups 1 to 13

[18] We also include an indicator for the Lifeline tariff and for errors updating the Lifeline tariff in the months following the meter switch.

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in Phase 1 and groups 14 to 27 in Phase 2), and interact each with an indicator for each month $t$ of the project:

$$\mathbf{Z}_{gt} = \sum_{s=1}^{6} \mathbf{1}[t = s] \times \left(g \cdot \mathbf{1}\left[g \in [1, 13]\right] + g \cdot \mathbf{1}\left[g \in [14, 27]\right]\right). \tag{2}$$

The result is two instruments representing splines in the randomization order for each of the six months of the project, for a total of 12 instruments. Each customer-month observation in the first stage is predicted by a single instrument:

$$prepaid_{igt} = \delta' \mathbf{Z}_{gt} + \tau_t + \eta_i + u_{igt}, \tag{3}$$

where $\delta$ is the vector of coefficients on the instruments. Other mechanical transformations of the randomized phase-in order present alternative instruments and deliver slightly different local average treatment effects. We test robustness to two of these alternative instruments in Section IIIA: group dummies interacted with month and a single linear group order variable interacted with month. We use the predicted prepaid variable, $\widehat{prepaid}_{igt}$, in the second stage:

$$y_{igt} = \alpha + \gamma \widehat{prepaid}_{igt} + \tau_t + \eta_i + \epsilon_{igt}, \tag{4}$$

where $\gamma$ is the effect of prepaid electricity metering on outcome $y_{igt}$.

Recovering the causal effect of prepaid metering requires that the randomized phase-in order is uncorrelated with unobserved time-varying factors that affect electricity consumption, conditional on customer and month fixed effects. Customer and month fixed effects control for customer-level, time-invariant determinants of demand and month-specific factors that affect all customers, respectively. Online Appendix Table A.1 shows the correlation between preproject customer characteristics and the randomized group order (column 1), the two splines in group order (columns 2 and 3), the actual switch date (column 4), and whether the customer was ever switched (column 5). The small number of groups included in the randomization results in a lack of balance on a few characteristics in the cross section, however, these time invariant characteristics are absorbed in the customer fixed effects used in all analyses. To examine time-varying correlates of the randomization order, we plot preproject residuals after removing customer and month fixed effects from the relationship between average daily kWh and month, prior to the project. We plot the average residual within each randomization group for the 34 months (beginning in January 2012) leading up to the start of the project and observe relatively flat trends that do not differ substantially across groups (see online Appendix Figure A.6). Online Appendix Figure A.7 plots monthly average daily kWh for the project sample and a comparison group of postpaid customers elsewhere in the city, and shows parallel trends in the months preceding the project start.

Figure 1 highlights an important feature of the randomized phase-in design: the relatively short time window from November 2014 to April 2015 during which the experimental variation can be used to identify treatment effects. The analysis above is therefore restricted to this short time window. Nevertheless, we investigate how customer behavior evolves following the meter replacement using an event-study

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specification that identifies a separate coefficient, $\beta_e$, on each event-time indicator $E_{it}$ for $A$ months before and after meter replacement:

$$y_{igt} = \alpha + \sum_{e=-A}^{A} \beta_e \mathbf{1}[E_{it} = e] + \tau_t + \eta_i + \epsilon_{igt}. \tag{5}$$

We first estimate a version of (5) using the randomized meter phase in. Because meters were phased in over six months, our instruments $\mathbf{Z}_{gt}$ only allow us to identify $\beta_e$ up to four months before and after the switch. Although the phase-in period only lasted for six months, we observe electricity use over a longer time period. We can thus exploit panel variation in consumption to assess the longer-term impacts of prepaid meters. Following (5), we assess patterns in electricity use up to a year before and after the meter replacement, using OLS and restricting the sample to compliers to avoid estimating impacts off of variation in compliance in months outside of the phase-in window.[19]

## III. Results

We start by showing the impact of switching to a prepaid meter on electricity use. Next, we analyze persistence and dynamics effects. Finally, we turn to heterogeneity in the treatment effects by customer characteristics and behavior prior to the replacement project.

### A. *Electricity Use*

Figure 2 plots the median daily average kWh in each month, from January 2012 through June 2016, and the share of targeted customers switched to prepaid metering. The figure shows the strong seasonal pattern of consumption, which peaks in the South African winter, when many customers use electricity to heat their homes. The drop in usage that coincides with the prepaid metering project is clearly visible in the figure, providing suggestive time series evidence that prepaid meters may reduce electricity use.

We next exploit the experimental variation in the timing of the prepaid meter installations to test whether the variation shown in Figure 2 is caused by the meter replacement project. Table 2 shows the regression coefficients from estimating equation (1) using OLS (columns 1 and 3) and by estimating equation (4) using two-stage least squares (columns 2 and 4), with the outcome in levels (columns 1 and 2) and logs (columns 3 and 4). In the one to five months after customers receive a prepaid electricity meter, average daily usage falls by around 1.9 kWh per day, which corresponds to a decline of between 13.2 and 14.1 log points. The OLS and IV specifications are very similar in magnitude, with overlapping standard errors, suggesting that noncompliance introduces little bias in the estimated treatment

[19] Restricting the longer-run event study specification to compliers and estimating with OLS ensures that noncompliers are not identifying longer-run effects outside of the phase-in window. Event study regressions include both customer and calendar-time fixed effects alongside the event-time indicators; to allow separate identification of event- and calendar-time fixed effects, we omit both the month before the meter replacement and the first month in event time (i.e., both $e = -1$ and $e = -A$) (Borusyak and Jaravel 2017).

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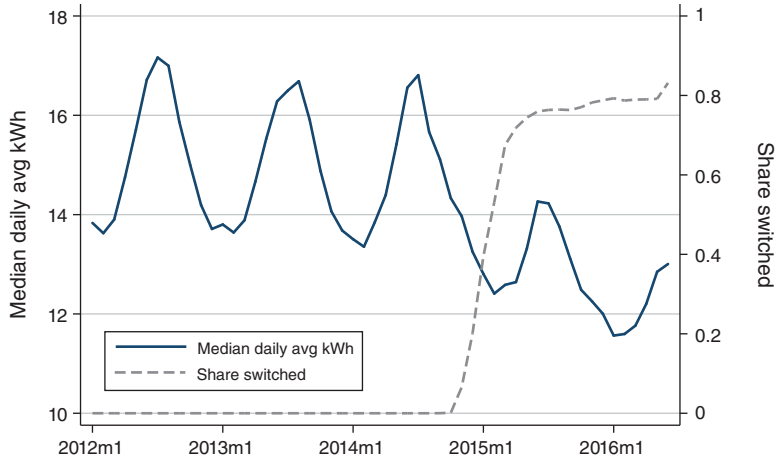


Figure 2. Daily Average kWh/Month and Share Switched over Time

*Notes:* Median daily average kWh in each month and the share of targeted customers switched to prepaid metering. Note that the peak months of June and July correspond to the winter months in Cape Town.

Table 2—Impact on Average Daily kWh

| | Avg daily kWh | | log(avg daily kWh) | |
|---|---|---|---|---|
| | OLS | IV | OLS | IV |
| | (1) | (2) | (3) | (4) |
| Prepaid | −1.908 | −1.923 | −0.132 | −0.141 |
| | (0.098) | (0.176) | (0.010) | (0.015) |
| Wild-bootstrap *p* | [0.000] | [0.000] | [0.000] | [0.000] |
| Observations | 21,759 | 21,759 | 21,663 | 21,663 |
| Number of customers | 4,175 | 4,175 | 4,168 | 4,168 |

*Notes:* The table shows the effect of prepaid metering on average daily kWh per customer per month in levels (columns 1 and 2) and logs (columns 3 and 4). Columns 1 and 3 report OLS coefficients on an indicator for prepaid metering. Columns 2 and 4 instrument for prepaid metering with the randomly assigned group order. See the main text for further description of the instrument. All specifications exclude the switch month, include a time-varying tariff control, and cluster standard errors at the randomization group level. All specifications include month (time) fixed effects and customer (household) fixed effects.

effect. Wild-bootstrap *p*-values are shown in brackets. The first stage for columns 2 and 4 is shown in online Appendix Table A.2.

*Dynamics and Persistence*.—We investigate dynamics using the event-study specification in equation (5), first restricting the window to the experimental phase-in period in panels A and B of Figure 3, which show results from OLS and IV regressions, respectively, and include noncompliers. Reductions in electricity usage are stable at around 2 kWh per day and significantly different from zero for each of the four months following the meter replacement. The magnitudes are similar in panels A and B, though estimated with less precision in the IV specification (panel B). Coefficients preceding meter replacement are never significantly different from zero

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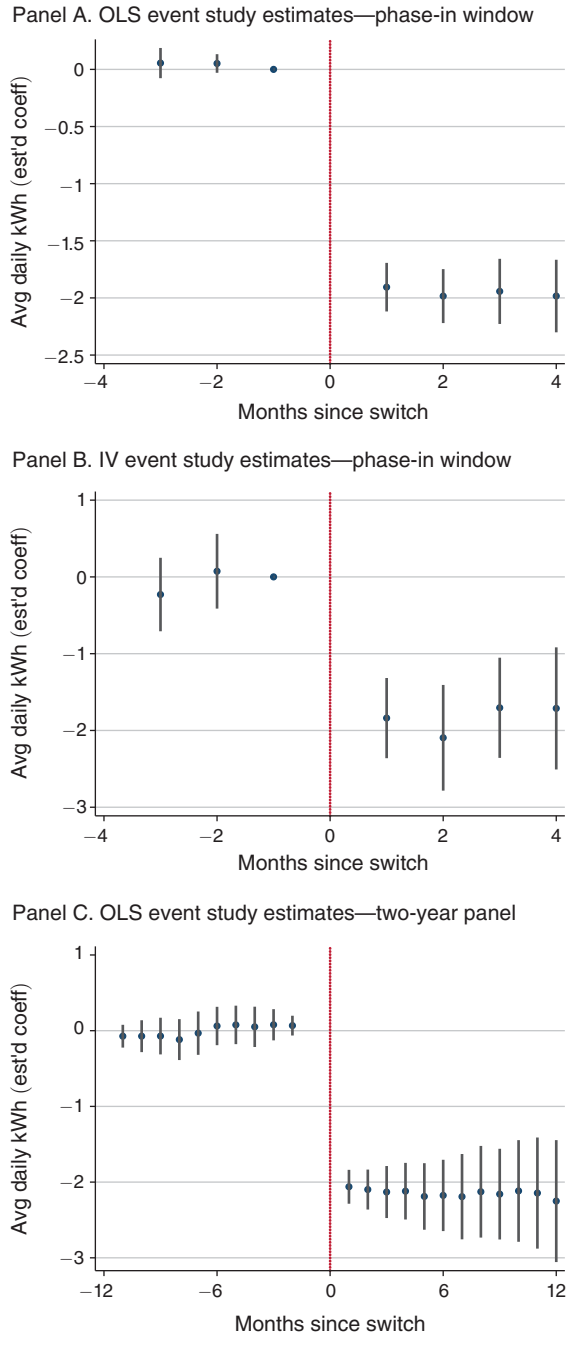


Figure 3. Event Study (Average Daily kWh)

*Notes:* Event study results for the effect of prepaid metering on average daily kWh. Panels A and B are limited to the phase-in window for all customers. Panel C includes up to one year before and after meter replacement and is limited to compliers. The IV specification (panel B) instruments for months since the metering switch with the randomly assigned group order. See the main text for further description of the instrument. All specifications include customer and time fixed effects and cluster standard errors at the randomization group level.

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in the months leading up to the meter replacement, further indicating little selection into the timing of or compliance with the meter replacement program.

Panel C extends the panel to up to a year before and after meter replacement, includes only compliers, and estimates the event-time coefficients using OLS. Analyzing effects over this longer window relies on less clean within customer variation, but shows no evidence of reversion to preproject levels of electricity usage (panel C). Up to one year following the meter replacement usage is persistently around 2 kWh below preproject levels.

*Robustness Checks*.—To examine the robustness of our estimate of the effect of prepaid electricity metering on electricity use, we re-estimate treatment effects on average kWh/day with alternative specifications, including alternative instruments and a difference-in-difference design that includes comparison customers not part of the meter replacement project, drawn from elsewhere in Cape Town to match the distribution of property values in the study sample.[20] As shown in online Appendix Table A.3, results are stable across the different specifications in columns 2–6. Online Appendix Figure A.7 shows the descriptive plot of electricity use underlying the main difference-in-difference specification in column 5.

Next, we examine the robustness to alternative sample restrictions and manipulations of the outcome variable. Online Appendix Table A.4 shows separate estimates for Phases 1 and 2 of the phase in, and online Appendix Table A.5 alternatively restricts the sample to different subgroups or alternative assumptions about the outcome.

Finally, we implement robustness checks designed to test remaining concerns about the implementation issues discussed in Section IIB; we exclude Lifeline customers who received the incorrect tariff when their meter was replaced and omit the two-thirds of customers on consolidated billing prior to the project. Results are unaffected by the tariff errors and are somewhat smaller for the one-third of customers not on consolidated billing (see online Appendix Table A.6). Note that customers receiving a separate electricity bill prior to the project use less electricity than the average customer and likely differ in numerous other ways; the smaller treatment effect in column 3 need not imply that the change from consolidated billing is an important part of our overall finding (see Section IIB for further discussion). Finally, we conduct a placebo test that reassigns the switch date and the instrument to one year prior to its actual occurrence and observe a small and insignificant coefficient on the prepaid indicator (online Appendix Table A.6, column 4), indicating that seasonal variation or other time trends are unlikely to drive the result.

[20] Note that neither of the comparison samples are ideal for difference-in-difference analysis. The sample of postpaid comparison customers (column 5) comes from other parts of the city since most postpaid customers in the project neighborhoods were targeted for meter replacement. The sample of prepaid comparison customers (column 6) is drawn from the same neighborhood but provides a different counterfactual: what consumption would have been if project customers had always been on prepaid metering.

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Table 3—Heterogeneous Treatment Effects

| | | | OLS | | IV | |
|---|---|---|---|---|---|---|
| | Number of customers | Mean | Total effects (1) | Difference (2) | Total effects (3) | Difference (4) |
| Domestic | 2,872 | 17.34 | −2.002 | 1.322 | −2.000 | 1.390 |
| | | | (0.121) | (0.241) | (0.172) | (0.337) |
| Lifeline | 1,303 | 9.74 | −0.680 | [0.000] | −0.609 | [0.000] |
| | | | (0.187) | | (0.290) | |
| Above median kWh | 2,118 | 19.86 | −2.308 | 1.313 | −2.341 | 1.443 |
| | | | (0.113) | (0.144) | (0.221) | (0.272) |
| Below median kWh | 2,057 | 9.93 | −0.995 | [0.000] | −0.899 | [0.000] |
| | | | (0.118) | | (0.179) | |
| High prop value | 1,468 | 17.29 | −1.669 | −0.376 | −1.414 | −0.866 |
| | | | (0.168) | (0.192) | (0.354) | (0.447) |
| Low prop value | 2,707 | 13.71 | −2.045 | [0.100] | −2.280 | [0.000] |
| | | | (0.095) | | (0.210) | |
| Usually on time | 2,091 | 13.97 | −1.576 | −0.650 | −1.751 | −0.311 |
| | | | (0.103) | (0.126) | (0.184) | (0.263) |
| Usually late | 2,084 | 15.97 | −2.225 | [0.000] | −2.062 | [0.600] |
| | | | (0.126) | | (0.248) | |
| No debts | 3,100 | 15.02 | −1.779 | −0.460 | −1.881 | −0.164 |
| | | | (0.098) | (0.176) | (0.189) | (0.438) |
| Outstanding debts | 1,075 | 14.80 | −2.240 | [0.000] | −2.045 | [0.700] |
| | | | (0.174) | | (0.417) | |
| Never disconnected | 3,334 | 14.88 | −1.890 | −0.097 | −1.912 | 0.031 |
| | | | (0.101) | (0.306) | (0.183) | (0.522) |
| Ever disconnected | 841 | 15.32 | −1.987 | [0.700] | −1.881 | [0.900] |
| | | | (0.284) | | (0.491) | |

*Notes:* Effects of the prepaid meter on log average daily kWh by subgroup. Columns 1 and 2 report OLS coefficients and columns 3 and 4 report IV coefficients. Wild bootstrap $p$-values are shown for the interaction term (difference) and reported in square brackets (columns 2 and 4). Specifications include time fixed effects interacted with the heterogeneity variable. All characteristics are defined by the preproject period as follows: "Lifeline" indicates customers primarily on Lifeline tariff, "Low prop value" indicates a 2012 ZAR property value below 300,000, "Usually late" indicates an above median probability of paying monthly bills past the due date, "Outstanding debts" indicates multiple unpaid bills at the end of the panel. "Ever disconnected" equals one if the customer was ever disconnected on postpaid metering.

## B. *Heterogeneous Treatment Effects*

We preview the heterogeneity that we will return to in the cost-benefit analysis (Section V), by examining impacts on use for different types of customers. We re-estimate equation (4), interacting each binary heterogeneity variable with the prepaid indicator and with each of the instruments.[21] Given the substantial differences in baseline usage by group, we report the postpaid average daily kWh for each subgroup in the column labeled mean and show results with the outcome in logs in online Appendix Table A.7. Columns 1 and 2 of Table 3 show the OLS effect on average daily kWh and columns 3 and 4 show IV estimates. The total effect for each subgroup is shown in columns 1 and 3 and the difference (i.e., the coefficient on the interaction term) in columns 2 and 4, along with wild-bootstrap $p$-values for the

[21] We also interact the time fixed effects and tariff error indicator with the heterogeneity variable to allow for differential time trends by group.

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estimated difference. The pairwise correlations between the heterogeneity variables is shown in online Appendix Table A.8.

We observe some pronounced heterogeneity in response to prepaid metering by subgroup and focus on the IV results in columns 3 and 4 in describing the results. Not surprisingly, Lifeline customers, who consume substantially less power on postpaid metering, also reduce their electricity use less, both in levels and in proportion to their baseline usage;[22] a similar pattern is evident when the sample is split by high and low baseline usage. When we divide customers according to the property valuation used for many other antipoverty programs in the City of Cape Town (300,000 ZAR based on the 2012 valuation, which corresponds to around US$[2014]29,000), we see larger reductions among low-property-value customers, both in proportion to their average consumption on postpaid metering and in levels. Differences by measures of delinquency on postpaid metering are not significant in the IV specification, though customers who usually pay their bills late and have outstanding debts reduce their usage by (insignificantly) more than less delinquent customers.

## IV. Mechanisms: Purchasing Patterns and Price Feedback

The switch from postpaid to prepaid metering changes numerous aspects of the customer experience. Our main treatment effects are therefore the consequence of what might be viewed as a bundled intervention. A list of potential mechanisms (which is likely not exhaustive) includes (i) increased enforcement of the relationship between payment and consumption; (ii) flexibility around the timing and frequency of expenditures; (iii) higher transaction costs; (iv) more information about or higher salience of consumption, expenditures, and prices; and (v) a technology to enable self control and intrahousehold control. A full accounting of the relative importance of each of these channels requires a different experimental design and considerably more data than are available from this study. Instead of attempting such an exercise, we focus on one potentially important channel: price information or salience.

Switching from postpaid metering to prepaid metering draws attention to the marginal price of electricity as follows. On a monthly bill, customers receive information every 30 days or so, describing their total consumption and the total amount owed along with details on the marginal price and amount owed on each tariff block. Of course, customers may be inattentive to this breakdown, and the act of purchasing electricity in advance of consumption may draw attention to prices or the tariff schedule even holding fixed the information content.[23] However, price feedback on a prepaid meter is also affected by how customers are charged for electricity over the

[22] Note that the Lifeline sample includes customers who were erroneously switched to the Domestic tariff. The fact that they are less responsive to the meter replacement than Domestic customers (all of whose tariffs remained the same) suggests that the change in tariffs is unlikely to explain the reduction in usage.

[23] In addition, as discussed in Section IB, some customers in our sample received a consolidated bill for electricity, water, and other municipal services, which may further reduce attention to electricity pricing, which is provided in a breakdown of the bill total by service category. In online Appendix Table A.6, we show robustness of our main results to focusing on customers who receive separate electricity bills when on postpaid metering. The treatment effect is smaller, though—as discussed in Sections IIB and IIIA—this may be due to selection into the different billing regimes.

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course of the month, and how often they choose to purchase prepaid electricity. As discussed in Section IB, customers on prepaid metering move up the tariff schedule based on cumulative purchases during the month. Thus, a customer who purchases multiple times during the month on a nonlinear price schedule receives different amounts of electricity for the same monetary expenditure at different times during the month. While customers could replicate their expenditure patterns on a monthly bill by purchasing prepaid electricity once a month, which would hold information about total expenditures and average price constant, customers in our sample purchase prepaid electricity every 3.1 days on average.[24] As a result, even if customers remain inattentive to their marginal price or to the structure of the tariff schedule, the average price associated with a given purchase will be closer to the marginal price the more frequently customers purchase electricity (online Appendix Figure A.8 plots average and marginal prices in the prepaid transaction data at the monthly level and per transaction).[25] Specifically, customers who are inattentive to the tariff schedule still receive information that they have crossed to a higher marginal price step at the time of purchase. If customers respond to this higher marginal price by curtailing further electricity purchases that month, we expect to observe bunching to the right of the tariff step. We exploit the nonlinear tariff to analyze the relevance of this form of price feedback for observed demand by (i) providing visual evidence of bunching at the tariff kink points, and (ii) quantifying the magnitude of the consumption response.

First, we generate histograms of monthly usage based on total imputed daily usage over the month.[26] We split the sample by tariff and meter type, where the variation in meter type comes from the replacement project (i.e., before and after), and restrict the analysis to compliers (i.e., customers who switched to prepaid metering under the project). Specifically, we use postpaid data from the 2013 billing year (July 2013 to June 2014) before the start of the project, and data from the 2015 billing year (July 2015 to June 2016) when all compliers have been switched to prepaid metering. Figure 4 shows a histogram of consumption around the relevant tariff step (full histograms are provided in online Appendix Figure A.9). Panel A of Figure 4 shows that Domestic tariff customers (left figure) exhibit no bunching on either postpaid or prepaid in response to the roughly 20 percent increase in marginal price at the tariff step at 600 kWh. Lifeline tariff customers (right figure), on the other hand, exhibit some modest bunching to the right of the nearly 250 percent increase

[24] This is similar to the purchase frequency documented in Jack and Smith (2015) for customers in the bottom half of the property value distribution throughout the City of Cape Town. As shown in online Appendix Table A.8, customers who purchase more frequently on a prepaid meter also consume more electricity on a postpaid meter, are more likely to pay their bills late, are more likely to face disconnections, and are poorer, as measured by their property value.

[25] The results that we describe below are therefore a combination of the endogenous purchase frequency and the price sensitivity of customers who choose to purchase at higher and lower frequencies.

[26] The use of imputed consumption data for this exercise raises the concern that we might miss some bunching that would be observed with actual consumption data. In addition, since customers on prepaid metering purchase electricity, potentially in round values, bunching might arise mechanically through expenditure patterns. However, we observe that most purchases are in round monetary values while we analyze bunching in monthly consumption quantities. For Lifeline customers (the focus of the bunching analysis below), the three most common purchase values (in ZAR) account for 72 percent of the transactions, while the three most common purchase quantities (in kWh) account for only 42 percent of the transactions.

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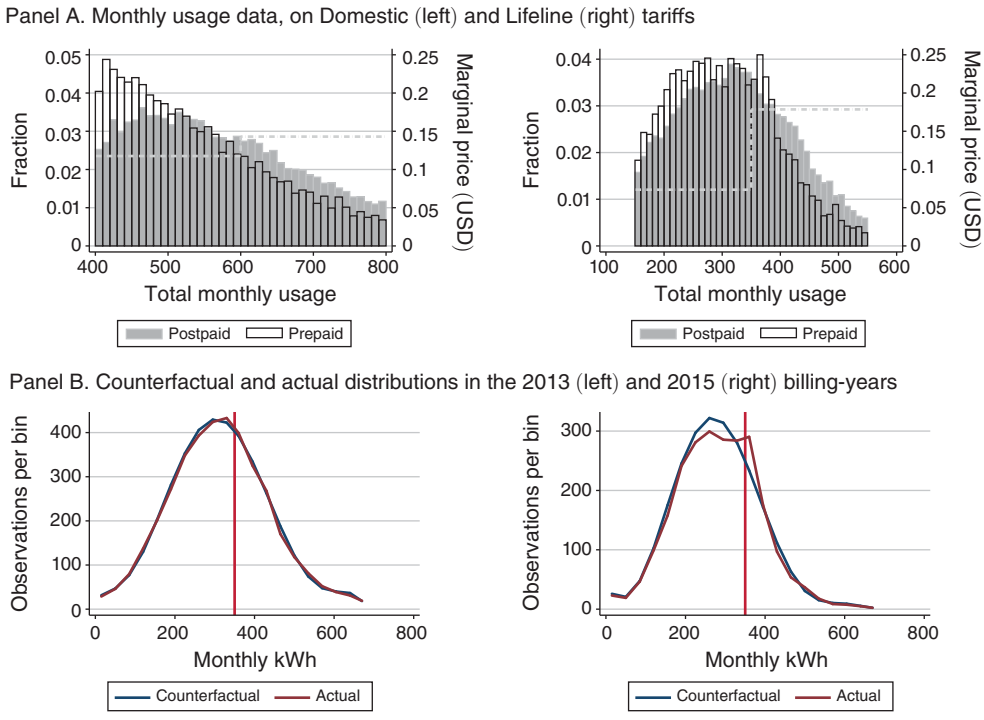


FIGURE 4. DISTRIBUTION OF MONTHLY ELECTRICITY USAGE

*Notes:* Histograms show customer-month observations of the imputed average daily kWh variable aggregated over the days in the month in 10 kWh bins, with tariff schedules overlaid (panel A). Panel B shows the actual and counterfactual distribution of customer-month observations in each 10 kWh bin restricted to Lifeline tariff customers whose meters were replaced as part of the project. Billing-years run from July to June so the 2013 (2015) billing-year covers July 2013 (2015) to June 2014 (2016). All meter replacements occurred between November 2014 and April 2015.

in marginal price at the tariff step at 350 kWh.[27] To quantify the magnitude of this behavioral response, we therefore restrict attention to observations on Lifeline tariff, where the variation in marginal price is much greater.[28] Note that bunching to the right of the tariff step is consistent with the price feedback mechanism discussed above; customers acquire information about the price through their purchases, and adjust behavior only after crossing the tariff step.

To quantify the magnitude of the response to the tariff kink, and how it differs by metering type, we follow a substantial recent literature on bunching in response to discontinuities in the slope of choice sets, here created by the change in the marginal price of electricity (see Kleven 2016 for an overview). Under a counterfactual constant marginal price of electricity, $p_1$, assume that consumers are distributed in their electricity use $E$ according to some smooth distribution $h_0(E)$. Introducing

[27] This is consistent with Chetty et al.'s (2011) finding that the amount of bunching is increasing in the magnitude of the kink, suggesting some fixed cost associated with re-optimizing behavior.

[28] This restriction excludes customers who received the Domestic tariff in error following the metering switch. These customers exhibit no bunching at 350 kWh after the switch to prepaid metering.

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a convex kink to the tariff schedule at quantity $E^*$ increases the marginal price to $p_2 = p_1 + \Delta p$. Consumers in some window $E^* + \Delta E^*$ prefer to consume $E^*$ following the change in marginal price. This response produces excess bunching at the tariff step. If consumers can perfectly optimize, then no consumer with $E < E^*$ or $E > E^* + \Delta E^*$ will adjust their behavior. The bunching is then defined by

$$B = \int_{E^*}^{E^*+\Delta E^*} h_0(E)\, dE \simeq h_0(E^*)\Delta E^*,$$

where the approximation is based on a counterfactual density that is constant on the bunching segment. Identifying bunching therefore requires (i) determining the bunching window $\Delta E^*$ and (ii) identifying the counterfactual distribution of electricity use in the absence of the price kink.

To allow for a flexible distribution of counterfactual usage, we follow Chetty et al. (2011) and fit a flexible polynomial to the observed distribution, excluding the observations in the bunching window $E^* + \Delta$. In the main specification, $\Delta$ is set to 40 kWh to the right of the tariff step, based on visual analysis of Figure 4 and the proposed mechanism, which predicts bunching to the right of the tariff step. Customer-month observations are grouped into electricity usage bins indexed by $j$, and the counterfactual is estimated using a regression of the number of observations in bin $c_j$ on a polynomial of degree $p$ in the bins:

$$c_j = \sum_{i=0}^{p} \beta_i \cdot (E_j)^i + \sum_{i=E^*}^{E^*+\Delta} \gamma_i \cdot \mathbf{1}[E_j = i] + \nu_j,$$

where bins are constructed in 10 kWh increments (the smallest typical purchase quantity), $E_j$ is the electricity use level in bin $j$, and $[E^*, E^* + \Delta]$ is the excluded range. We use a seventh order polynomial in our main specification. Panel B of Figure 4 plots the predicted counterfactual distribution and the actual distribution for compliers in the billing year before and after the meter replacement project (i.e., July 2013–June 2014 for postpaid metering and July 2015–June 2016 for prepaid metering).[29] The difference between the counterfactual and actual distribution around the tariff step at 350 kWh is considerably more pronounced on prepaid than on postpaid metering. Specifically, we estimate that the difference between the observed and the counterfactual mass in the 40 kWh window to the right of the tariff step is $\Delta B = -0.002$ (standard error 0.027) for postpaid observations and $\Delta B = 0.212$ (standard error 0.074) for prepaid observations. The $t$-statistic associated with the difference between these estimates is 2.70. Online Appendix B provides additional detail, along with robustness checks for alternative bunching windows and polynomial orders.

As Kleven (2016) and Blomquist and Newey (2017) make clear, using the estimated bunching mass to identify the marginal price elasticity faces two important identification challenges. First, in the presence of optimization

[29] We satisfy the integration constraint by iteratively increasing the counterfactual distribution by a uniform amount for each bin until the constraint is satisfied. An alternative approach that increases the counterfactual distribution proportional to the share of the observations that fall in each 10 kWh bin results in similar estimates, but a less close visual match to the histograms than the uniform approach.

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errors, $\Delta E^* = g(e, \phi)$, where $e$ is the price elasticity of demand and $\phi$ is the optimization error, $e$ and $\phi$ are not separately identified (Kleven 2016). Second, even if the counterfactual distribution $h_0$ is correctly specified, additional assumptions about preference heterogeneity are needed to identify an average elasticity parameter when preferences affect both the counterfactual distribution and the distribution of elasticities (Blomquist and Newey 2017). We focus on the former issue in interpreting our estimates and assume that the heterogeneity in counterfactual usage is uncorrelated with elasticities (within a narrow bunching window) and correctly approximated by $h_0$.

We use the estimated difference between the observed and counterfactual mass to the right of the 350 kWh tariff step ($\Delta B$) to calculate an implied marginal price elasticity, $\varepsilon = \%\Delta B / \%\Delta p$. For the observations on postpaid metering, the implied marginal price elasticity is 0.001 (standard error 0.021). For prepaid metering, it is $-0.121$ (standard error 0.042).[30] The $t$-statistic for the difference between these estimates is $-2.45$. Note, however, that this difference need not be interpreted as an actual change in marginal price elasticities; indeed, the explanation that we advance above instead suggests that it may be driven entirely by a reduction in optimization frictions. This interpretation is consistent with Ito's (2014) result that customers respond to average rather than marginal prices on postpaid metering, which would also explain why postpaid customers have finite demand given their estimated zero marginal price elasticity. In our setting, the change in metering type draws attention to marginal prices through the (endogenously) higher frequency price feedback on a prepaid meter.[31]

As a final step, we perform a simple calibration to evaluate the potential contribution of the change in price sensitivity to the roughly 14 percent drop in consumption shown in Table 2. Under the somewhat strong assumption that the change in sensitivity to marginal price from the bunching analysis can be generalized to other points along the tariff schedule and to customers whose demand falls outside of the bunching window, we calculate that up to 51 percent of the change in demand may be explained by the change in price sensitivity. We describe the calibration and additional assumptions in greater detail in online Appendix B.

## V. Costs and Benefits to the Utility

We calculate the costs and benefits of meter replacement for the City of Cape Town based on our empirical estimates and administrative cost records. We start by estimating impacts on other outcomes related to customer behavior, including revenue and cost measures. While the objective function of the municipal utility is almost certainly not profit maximization, comparing net revenue across metering

[30] A comparison across a random sample of prepaid and postpaid customers elsewhere in Cape Town shows qualitatively similar results when comparing across meter types in the same year.

[31] It is important to note that if customers ignore the tariff structure that underlies the price feedback, then the average price feedback that customers receive from purchases early in the month may be equal to their marginal price for that transaction, but does not fully reflect the opportunity cost of their purchase, which also includes the likelihood of moving to a higher price step later in the month. Thus, while we document an increase in sensitivity to the tariff structure, which allows customers to better respond to price, the nature of the feedback may distort consumption within the month.

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types is—we believe—relevant for the policy questions around expanding access that helped motivate this study. To this end, we also examine heterogeneity across customers in the returns to the utility from prepaid metering (relative to postpaid) and consider how our estimates generalize to other settings.

### A. *Revenue, Costs, and Timing*

Following the empirical strategy described in Section IID, we estimate several additional impacts of the meter replacement project. Results are shown in Table 4, which includes both OLS (panel A) and IV (panel B) specifications, following equations (1) and (4).

*Revenue Due and Revenue Forgone*.—Electricity is priced on an increasing block tariff, which means that reductions in consumption may lead to proportionally larger reductions in the amount due from customers whose reduction crosses a tariff step or proportionally smaller (zero) reductions for customers who only consume their free electricity allowance on either meter type. We estimate the impact of prepaid metering on revenue due for consumption in a month (Table 4, column 1). Revenue due falls by US$7.30/month in the OLS specification (panel A) and US$6.90/month in the IV specification (panel B). These results correspond to a reduction of around 13 percent each month relative to the average amount owed on postpaid metering (see Table 1).

The probability that the revenue due is ever paid is also impacted by the switch to prepaid metering. After three years, unpaid bills are written off by the municipal utility. We construct a measure of foregone revenue, which includes both unpaid bills that are more than three years old and an estimated payment probability for bills that remain unpaid at the end of our panel but are less than three years old.[32] The impact on foregone revenue is worth −5.9 US dollars per customer per month to the utility in the OLS specification (panel A) and −6.4 US dollars in the IV specification (panel B).

*Payment Timing*.—The customer response to prepaid metering consists of both the quantity of electricity consumed and the timing of payment. Mechanically, prepaid metering moves the payment date from after consumption to before consumption. In addition, the billing cycle adds a delay of around 60 days to the time between when a postpaid metering customer consumes and when payment is due. As discussed in Section IIA, our measure of payment delay is calculated as the time between when consumption occurs and when payment arrives to the utility, measured in months. As shown in column 3 of Table 4, the switch to prepaid metering reduces the time between consumption and payment by −3.1 months in both specifications.

We monetize the value of the change in payment timing by calculating the value of the associated interest, using the municipal utility's annual interest rate

[32] To calculate payment probabilities for unpaid bills at the end of our panel that are younger than three years, we calculate a "hazard rate," or the probability of default given that no payment has occurred by the number of months since billing. See Section IIA and online Appendix C for further discussion.

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Table 4—Revenue-Related Outcomes

| | Revenue due (1) | Foregone revenue (2) | Payment delay (3) | Foregone interest (4) | Supply cost (5) | Billing cost (6) |
|---|---|---|---|---|---|---|
| *Panel A. OLS* | | | | | | |
| Prepaid | −7.325 | −5.943 | −3.052 | −1.164 | −3.109 | −1.292 |
| | (0.374) | (0.452) | (0.133) | (0.056) | (0.153) | (0.189) |
| Wild-bootstrap *p* | [0.000] | [0.000] | [0.000] | [0.000] | [0.000] | [0.000] |
| *Panel B. IV* | | | | | | |
| Prepaid | −6.872 | −6.428 | −3.079 | −1.122 | −3.078 | −0.620 |
| | (0.716) | (0.766) | (0.147) | (0.054) | (0.305) | (0.434) |
| Wild-bootstrap *p* | [0.000] | [0.000] | [0.000] | [0.000] | [0.000] | [0.200] |
| Observations | 21,759 | 21,759 | 19,976 | 19,976 | 21,759 | 21,759 |
| Number of customers | 4,175 | 4,175 | 4,099 | 4,099 | 4,175 | 4,175 |

*Notes:* The table shows the effect of prepaid metering on the revenue due for consumption (column 1), the value of revenue foregone due to customer defaults (column 2), the months between consumption and payment (column 3), the value of the interest associated with the payment delay (column 4), the monthly supply cost (column 5), and the monthly billing cost (column 6). Columns 3 and 4 omit bills that are never paid. Specifications follow columns 1 and 2 in Table 2.

of 8 percent.[33] Results are shown in column 4 and imply a reduction in foregone interest of −\$1.10 US dollars per customer per month in both specifications. Note that the analysis of payment timing (columns 3 and 4) excludes payments that never arrive.

*Supply and Billing Costs*.—The utility purchases electricity from the national electricity company based on prices that vary with time of day and month of the year. While we do not observe the time of the day of consumption, we do observe the month of the year. In the experimental phase-in window, there is no variation in average marginal supply costs; these are higher in June, July, and August, suggesting larger savings following the end of our analysis. We analyze the impact on the cost of supply, accounting both for the average marginal supply cost and the fact that, on average, 11.25 percent of electricity supplied to the grid is lost.[34] Column 5 of Table 4 shows that the cost of supply falls by −\$3.10 US dollars per customer per month in both specifications.

Finally, we calculate the cost of revenue recovery by including a monthly measure of (i) disconnection and reconnection costs, which equal US\$90;[35] (ii) meter reading and bill preparation costs, which we assume to be the same for all postpaid

[33] The City of Cape Town uses a discount rate approved by the National Energy Regulator of South Africa (NERSA) for project evaluation. From 2010 to 2016, the rate was 8 percent, as stipulated by the South African electricity grid code.

[34] This number includes both technical and nontechnical losses. We assume that these are the same across meter types. Of course, the type of meter may also impact theft, which is classified as a nontechnical loss by the utility. They are not able to measure losses by meter type, nor do they have estimates of the impact of prepaid metering on electricity theft.

[35] We include the cost of both disconnection and reconnection in the month that a disconnection is observed in the data. Around 20 percent of the customers in our data are disconnected (and eventually reconnected) in at least one month while on postpaid metering. Conditional on ever being disconnected, the average number of disconnections is two. The total cost of a disconnection and reconnection is around US\$120, US\$30 of which is charged to the customer.

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customers in the data and average US$1.74 per customer per month;[36] and (iii) a fixed margin per kWh of US$0.002 applied to all electricity purchased through the prepaid vending system, including free electricity. These revenue recovery costs fall by −$1.30 US dollars per customer per month in the OLS specification (panel A) and by −$0.62 US dollars in the IV specification (panel B), which is not significantly different from zero.

### B. *Returns to the Municipal Utility from Meter Replacement*

We write down a simplified expression for the present value of revenue net of recurring costs under each metering type $m = \{pre, post\}$:[37]

$$PV^m = \frac{pq^m}{(1+i)^{t+s^m}} - \frac{cq^m + b^m}{(1+i)^t}. \tag{6}$$

Calculation of the monthly present value of net revenue draws on three parameters, which combine the outcomes analyzed in Table 4: (i) the revenue paid to the utility ($pq^m$), which combines columns 1 and 2; (ii) the timing of payment ($s^m$), as shown in column 3; and (iii) the cost of supply and revenue recovery ($cq^m + b^m$), which combines columns 5 and 6. Our calculations ignore fixed costs associated with infrastructure and operation, which are covered out of the rate structure in Cape Town. Thus, results should not be interpreted as a measure of overall utility profits; instead, they reflect the payoffs to the utility from adding a customer to either prepaid or postpaid metering.

*Relative Returns to the Utility*.—We calculate the value to the municipal electric utility of prepaid metering relative to postpaid metering (i.e., $PV^{pre}/PV^{post}$) by plugging our estimates from the meter replacement project and administrative cost numbers obtained from the City of Cape Town into (6). A summary of the values we use is provided in Table 5, which shows predicted means under each metering type, following the IV specification and measurement in panel B of Table 4. The final row shows the average per-customer monthly present value of net revenue for the City of Cape Town. Bootstrapped standard errors are in parentheses. The results show that the average customer generates around 23 percent higher net revenue on prepaid metering than on monthly postpaid billing. Note that this proportional gain ignores the one-time cost of the meter and so is constant over time (i.e., the relative returns from prepaid metering is 23 percent higher over any time horizon). It takes less than a year for the additional net revenue (US$50.40 per year) to cover the additional fixed cost of the prepaid meter (US$66, or around US$33 more than a new postpaid meter).

[36] Monthly billing costs vary across customers. We use a system-wide average for residential customers which may be below the cost for customers in our sample who tend to be located in areas with a relatively low density of postpaid meters that require monthly readings. Using this number also leads to a conservative estimate of the cost savings from switching customers to prepaid meters if entire meter reading routes are eliminated.

[37] These calculations avoid a more complete modeling of consumer and utility decisions. For example, utility decisions about enforcing nonpayment affect both revenue recovery rates and revenue recovery costs. On the customer side, we take point estimates from our impact evaluation as given and do not model the mechanisms that relate the metering technology to consumption, including differences in price elasticities of demand, suggested by the analysis in Section IV. In addition, we ignore capital and fixed supply costs, including the meter itself.

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Table 5—Present Value of Net Revenue ($i = 0.08$)

| | Postpaid | Prepaid |
|---|---|---|
| *Revenue estimate* | | |
| Revenue ($pq^m$) | 45.98 | 45.54 |
| | (0.32) | (0.59) |
| *Payment timing estimate* | | |
| Months since consumption ($s^m$) | 2.67 | −0.41 |
| | (0.06) | (0.09) |
| *Cost estimate* | | |
| Cost ($cq^m + b^m$) | 26.77 | 23.07 |
| | (0.18) | (0.32) |
| Includes: | | |
| Meter reading | 0.91 | 0 |
| Billing | 0.85 | 0 |
| Vendor commission | 0 | 0.002/kWh |
| Losses (percent) | 11.25 | 11.25 |
| Disconnections | 90 | 0 |
| Present value | 18.31 | 22.44 |
| | (0.39) | (0.56) |

*Notes:* Monthly net revenue in USD 2014 by metering type for the City of Cape Town. Predicted means and standard errors are reported for each estimate, based on the specification in column 2 of Table 2. The bottom panel presents the present value of net revenue per month at an annual interest rate of 8 percent, with bootstrapped standard errors.

Table 4 is useful for disaggregating this summary measure. Focusing on panel B, the difference between revenue due (column 1) and cost of supply (column 5) is a decrease in net revenue of around US$3.80 per customer per month. This gap is more than made up for by reductions in foregone revenue (column 2), foregone interest (column 4), and lower revenue recovery costs (column 6). Over half of the total savings to the municipal utility (inclusive of electricity supply costs) come from more complete revenue recovery (column 2). Lower supply costs (column 5) contribute around a third of the savings, with the value of earlier payments (column 4) and lower billing and revenue recovery costs (column 6) contributing around 10 and 5 percent, respectively. This breakdown highlights the value of more complete revenue recovery, even in a setting where eventual payment probabilities are high relative to many developing countries.

The analysis in this section highlights two other important aspects of revenue recovery under prepaid metering. First, the relative returns from prepaid metering are considerably below what would be expected if the reduction in consumption were ignored. If consumption did not change, then the difference in administrative costs, together with earlier payments, leads to payoffs that are almost 50 percent higher than the number we calculate. Second, for monthly consumption less than 880 kWh per month (93 percent of customer-months on postpaid), the costs of prepaid vending commissions are less than the costs of meter reading and bill preparation. This implies that customers with low levels of consumption are particularly costly to serve on a postpaid model relative to a prepaid model, even if all bills are paid on time. We turn next to understanding how this and other sources of customer heterogeneity impact the utility's returns from prepaid metering.

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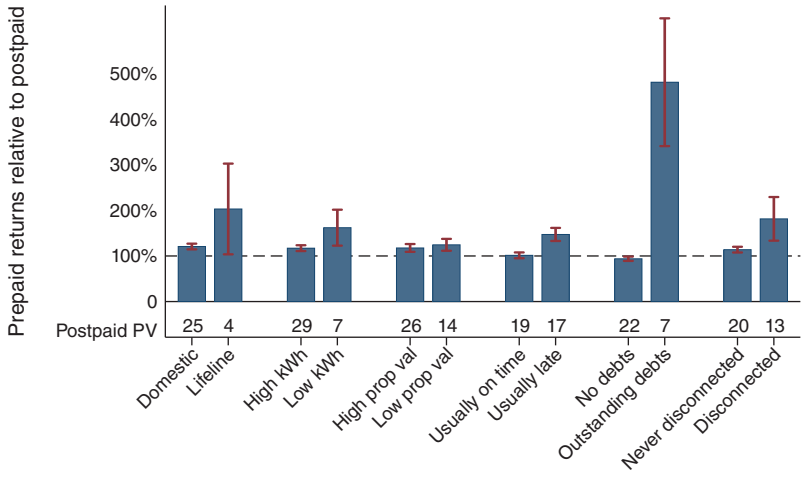


Figure 5. Heterogeneity in Relative Returns to Prepaid Metering

*Notes:* Returns to prepaid metering relative to postpaid metering per customer per month. Separate effects are calculated for each customer characteristic (each pair of bars splits the population), and standard errors are bootstrapped. Heterogeneity variables are as described in Table 3. The present value of the monthly returns to the utility for each customer type on postpaid metering are printed below the bars.

## C. *Heterogeneity and Generalizability*

The costs and benefits of prepaid metering depends both on the consumption and payment behavior of the customer base and also on features of the tariff and cost environment. As a result, the numbers from Cape Town need not generalize to other settings. We extend our average cost benefit calculation to examine how the results vary by customer type and the administrative setting.

First, we revisit the customer characteristics analyzed in Section IIIB. We calculate estimates of the returns from prepaid metering relative to postpaid metering for each subgroup. Note that a relative return of 100 percent implies that prepaid and postpaid meters generate the same returns to the utility. Figure 5 summarizes the results along with bootstrapped 95 percent confidence intervals (see also online Appendix Table A.9). We include the monthly net revenue from postpaid metering for each customer type on the figure, and observe considerable heterogeneity by characteristic in the baseline returns to the electric utility for serving each type of customer. Prepaid metering improves returns for customers on both tariff schedules, though the effects are larger (with wider confidence intervals) for customers on the subsidized Lifeline tariff who are less profitable to begin with. Low consumers generate higher relative returns than high consumers, again starting from a less profitable average present value on postpaid metering. Returns are similar across property values, with a slightly higher estimated impact on the returns from lower property value customers. Perhaps unsurprisingly, relative returns improve when delinquent customers are switched to prepaid (i.e., those with late payments, outstanding debts, or a history of disconnection on postpaid). Less delinquent customers, including those who usually pay on time and those with no outstanding payments at the end of the panel, actually generate very slightly negative returns on prepaid relative to postpaid metering. Notably, the customer characteristics

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that are associated with the greatest improvements in net revenue under prepaid metering are likely to be considerably more prevalent in other developing country settings; they are the types of customers most likely to be excluded from service on a postpaid billing system.

Next, we look to alternative assumptions about the cost environment. Online Appendix Figure A.10 shows basic simulations to support the intuition that we lay out here. First, in Cape Town, the discount rate used for project evaluation is relatively low at 8 percent. This places little weight on the value of revenue received earlier under prepaid metering, and higher interest rates are associated with a higher relative value for prepaid metering. Second, technical and nontechnical losses on the City of Cape Town's grid are around 11.25 percent. Utilities in other developing countries tend to have higher loss rates and may face a higher cost of supply, resulting in higher relative payoffs from prepaid metering, assuming losses are unaffected by the type of metering. The intuition is similar if tariffs are below marginal cost, which Trimble et al. (2016) show is the case for the vast majority of countries in sub-Saharan Africa: reductions in consumption are less costly to the utility as the marginal kWh becomes less profitable.[38] Third, the assumption that nontechnical losses—which include illegal consumption—are unaffected by the type of meter may not hold. We lack the data to investigate whether the decrease in legal consumption that we document is partially offset by an increase in illegal consumption. A sufficiently large increase in illegal consumption could negate all of the net revenue gains to the utility. In our setting, combined technical and nontechnical losses—we cannot disentangle these in our data—would have to nearly triple before the switch to prepaid metering would represent a net loss to the utility.

## VI. Conclusion

Finding ways to expand energy access while maintaining the financial viability of the electricity sector presents a policy challenge in many developing countries. Across sub-Saharan African countries, the annual value of uncollected bills averages 0.17 percent of national GDP (Kojima and Trimble 2016). Prepaid metering has been proposed as a technological solution to improve revenue recovery at relatively low cost, yet claims have so far been largely anecdotal. We study the impact of prepaid electricity metering on residential electricity use and utility revenue and costs in Cape Town, South Africa. Using a unique dataset that tracks customers as they are switched from postpaid to prepaid metering, we document a persistent 14 percent decrease in electricity use following the switch. We show that at least some of the reduction in consumption appears to stem from the (endogenously) more frequent price feedback associated with prepaid electricity purchases, which increases customer sensitivity to the marginal price.

Customer responses, both in terms of overall consumption and in the effect of the metering change on payment patterns, have implications for utility revenue and cost

[38] Trimble et al. (2016) include several cost categories in their analysis that are excluded from our estimates because they do not depend on metering type or on small changes in customer demand. Thus, the net revenue that we estimate—per customer or per kWh—should not be directly compared with their numbers.

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flows. We calculate the costs and benefits to the utility from prepaid metering, relative to postpaid metering. We show that the relative returns from prepaid metering will depend on features of the cost environment, as well as on the customer base. In our sample, customers who consume less electricity and who are more delinquent on average yield the highest returns to prepaid metering relative to monthly billing. While the generalizability of our point estimates may be limited, this heterogeneity offers some insight into the potential returns to prepaid metering in other settings. In particular, the characteristics that we find to be predictive of higher relative returns to prepaid metering are likely to be even more common in other developing country settings. Numerous caveats to generalization are warranted; we focus on two. First, we document treatment effects among experienced electricity consumers. Newly connected customers may respond differently, and the effect on consumption may be larger or smaller than what we observe in Cape Town. Depending on tariffs and marginal costs, this has the potential to undo the benefits of prepaid metering for the electric utility. Second, we are unable to measure electricity theft on either postpaid or prepaid metering. Theft levels may be higher or lower on prepaid metering than on postpaid billing. A large enough increase in electricity theft may also undo the benefits of prepaid metering for the electric utility.

While our data do not allow for a clear accounting of the mechanisms underlying the consumption response to prepaid metering, we clearly show that impacts on consumption must be taken into consideration when assessing the utility's payoffs from prepaid metering. Ignoring them results in a substantial overestimate of the benefits of prepaid metering to the utility. A full welfare accounting must, of course, consider why customer electricity use changes. The impacts we document are substantial, and the welfare implications depend on whether they reflect conscious conservation enabled by greater control and improved information or salience, or whether they reflect constraints that leave households sitting in the dark at least some of the time. We leave a complete investigation of these potential channels underlying our results for future research, but note that in many settings, the choice between prepaid and postpaid metering is likely to affect the extensive margin of access. In these settings, the appropriate welfare comparison may be electrification with prepaid metering or no electricity access at all.

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